Project Documentation

Human Development Index (HDI) Prediction System — Final Report

1. Abstract

This project implements a Human Development Index (HDI) Prediction System that estimates a country's HDI score and classifies it into the official UNDP development tier using four socio-economic indicators: life expectancy at birth, expected years of schooling, mean years of schooling, and Gross National Income (GNI) per capita. The system is delivered as a Flask-based web application backed by a trained regression model.

2. Introduction

The Human Development Index is UNDP's summary measure of average achievement in health, education, and standard of living. While UNDP publishes official HDI values annually, there is no lightweight interactive tool for estimating HDI from custom indicator values. This project bridges that gap with a machine learning model wrapped in a simple web interface.

3. Problem Statement

Given a set of core development indicators for a hypothetical or real country, there is no simple way for a non-technical user to obtain an instant HDI estimate and its corresponding development tier without manually applying UNDP's normalization formulas.

4. Objectives

- Train a regression model that predicts HDI from four core indicators.
- Classify the predicted score into UNDP's four development tiers.
- Expose the model through an easy-to-use web form and a JSON API.
- Validate input and handle errors gracefully.

5. Dataset

The project uses the UNDP Human Development Index dataset (Dataset/HDI.csv), covering 195+ countries and territories with indicators spanning 1990 to 2021. Beyond the core HDI components, the dataset also includes gender-disaggregated indices, inequality-adjusted HDI, Gender Inequality Index, and environmental indicators such as CO2 emissions per capita — though only the four core indicators below are used as model features.

Feature	Column in Dataset	Role
Life Expectancy at Birth	Life Expectancy at Birth (year)	Input feature
Expected Years of Schooling	Expected Years of Schooling (year)	Input feature
Mean Years of Schooling	Mean Years of Schooling (year)	Input feature
GNI per Capita	Gross National Income Per Capita (year)	Input feature

Feature	Column in Dataset	Role
Human Development Index	Human Development Index (year)	Target variable

6. System Architecture

The system separates offline training from online serving. The Jupyter notebook (Training/Human Development Index.ipynb) loads HDI.csv, selects the four indicators, trains a regression model, and serializes it as HDI.pkl. The Flask app (Code/app.py) loads this model once at startup and serves predictions through the **/** and **/predict** routes, rendering results via templates/index.html and static/ assets for styling and interactivity.

7. Methodology

- Data Cleaning: rows/columns with excessive missing values for the selected features and target year were filtered out.
- Feature Selection: life expectancy, expected years of schooling, mean years of schooling, and GNI per capita were chosen as they directly correspond to UNDP's three HDI dimensions.
- Model Training: a regression model was trained (scikit-learn) to map the four features to the HDI value, then serialized with pickle.
- Tier Classification: predictions are clipped to [0, 1] and mapped to UNDP's official tiers using fixed thresholds (0.800 / 0.700 / 0.550) inside classify_hdi().

8. Implementation Details

The Flask application defines a **run_prediction()** helper that parses the four input fields, builds a NumPy feature array, calls **model.predict()**, clips the result to a valid HDI range, and classifies it via **classify_hdi()**. The **/predict** route supports both JSON requests (returning a JSON payload) and standard HTML form submissions (returning a rendered page with the prediction text), with error handling for missing or invalid values in both cases.

9. Testing Summary

The system was validated against test cases covering typical inputs across all four development tiers, boundary values at the tier thresholds, missing/invalid input handling, and both the form-based and JSON API request paths. All test cases passed; see Project Testing / Test Cases and Report.pdf for full details.

10. Results

The trained model produces HDI estimates consistent with the expected development tier for representative indicator combinations (e.g., high life expectancy, high schooling years, and high GNI per capita reliably map to the 'Very High' tier, matching UNDP's classification philosophy).

11. Conclusion & Future Scope

The HDI Prediction System successfully demonstrates how a small set of socio-economic indicators can be used to approximate a country's Human Development Index and development tier through a simple, accessible web interface. Future enhancements could include: predicting the

Inequality-adjusted HDI (IHDI) using the coefficient-of-inequality columns, adding year-over-year trend forecasting, and visualizing a country's position relative to regional peers.

12. References

- UNDP Human Development Reports – Human Development Index methodology.
- Project dataset: Dataset/HDI.csv (UNDP HDI composite indices, 1990–2021).
- Flask documentation – flask.palletsprojects.com
- scikit-learn documentation – scikit-learn.org